

5 - First-Order Logic and Knowledge Representation

Introduction to First-Order Logic

First-order logic (FOL), also known as first-order predicate logic, is a fundamental formal system used in mathematics, philosophy, computer science, and linguistics for expressing and reasoning about relationships between objects in a domain. In artificial intelligence (AI), first-order logic (FOL) serves as a cornerstone for representing and reasoning about knowledge. Its syntax and semantics provide a robust framework for encoding information in a precise and structured manner, enabling AI systems to perform tasks such as automated reasoning, planning, and natural language understanding.

Syntax of First-Order Logic

The syntax of first-order logic (FOL) is about the symbols used and the rules for combining them to make correct and meaningful statements.

These correct statements are called well-formed formulas (WFFs).

It also tells us how to write statements correctly using those symbols.

Terms in First-Order Logic

Terms represent objects or entities within the domain of discourse. In AI, terms can correspond to real-world entities, such as objects, individuals, or abstract concepts. They include:

Constants: Specific entities, e.g., "John", "Apple".

Variables: Placeholders for entities, e.g., "x", "y".

Functions: Expressions applied to terms, e.g., "Age(John)", "Parent(x)".

Semantics of First-Order Logic

Semantics in first-order logic deals with the interpretation of sentences and formulas within the framework of a mathematical model. It provides a way to assign meanings to the symbols and structures used in first-order logic.

Key Elements of Semantics in First-Order Logic

Variables: These represent placeholders for objects or elements within a domain.

Constants: These represent specific elements within the domain.

Predicates: These are expressions that can be true or false depending on the objects they're applied to.

Functions: These map elements from the domain to other elements in the domain.

Quantifiers: Such as "for all" ($\forall$) and "exists" ($\exists$), used to express universal and existential quantification, respectively.

Inference in First-Order Logic (FOL)

Inference in First-Order Logic is the process of **deriving new facts or conclusions** from known facts and rules using **logical reasoning**.

Knowledge Base (KB) in FOL

A **knowledge base** in FOL contains:

- **Facts** (specific information)
- **Rules** (general statements using variables)

Forward Chaining

It is a data-driven inference technique. It starts with the available data and applies rules to infer new data until a goal is reached. This method is commonly used in situations where the initial data set is extensive, and the goal is to derive conclusions from it.

How Forward Chaining Works

Start with Known Facts: The inference engine begins with the known facts in the knowledge base.

Apply Rules: It looks for rules whose conditions are satisfied by the known facts.

Infer New Facts: When a rule is applied, new facts are inferred and added to the knowledge base.

Repeat: This process is repeated until no more rules can be applied or a specified goal is achieved

Backward Chaining

Backward chaining is a goal-driven inference technique. It starts with the goal and works backward to determine which facts must be true to achieve that goal. This method is ideal for situations where the goal is clearly defined, and the path to reach it needs to be established.

How Backward Chaining Works

Start with a Goal: The inference engine begins with the goal or hypothesis it wants to prove.

Identify Rules: It looks for rules that could conclude the goal.

Check Conditions: For each rule, it checks if the conditions are met, which may involve proving additional sub-goals.

Recursive Process: This process is recursive, working backward through the rule set until the initial facts are reached or the goal is deemed unattainable.

Comparison of Forward and Backward Chaining		
Feature	Forward Chaining	Backward Chaining
Approach	Data-driven	Goal-driven
Starting Point	Known facts	Specific goals
Efficiency	Efficient for exploring all possible inferences	Efficient for achieving specific goals
Memory Usage	Can be memory intensive due to storing intermediate facts	Typically requires less memory
Implementation Complexity	Simple to implement	More complex to implement due to recursive nature
Suitability	Suitable for dynamic environments with continuously arriving data	Suitable for diagnostic systems and interactive applications
Handling Large Rule Sets	May become slow with large rule sets	Can be complex to manage with large rule sets

Knowledge Representation Beyond Classical Logic

While classical logic (propositional and first-order) is precise, it can be **too rigid** for real-world knowledge. AI uses **more flexible techniques** to represent knowledge:

A. Semantic Networks

Definition:

- A **graph-based representation** where **nodes** represent concepts or entities, and **edges** represent relationships.

Features:

- Easy to visualize.
- Supports inheritance (properties passed from general to specific nodes).

Example:

```
[Bird] ---isA---> [Animal]
[Penguin] ---isA---> [Bird]
[Penguin] ---cannotFly---> true
```

B. Frames

Definition:

- **Structured representation** like a table or object with **slots and values**.
- Each frame represents a **concept** and its **properties (slots)**.

Frame: Car	Slots	Values
Engine	Type	V6
Wheels	Number	4
Fuel	Type	Petrol

Scripts

Definition:

- A **structured representation of events or sequences**, capturing **typical scenarios**.

Example: “Restaurant Script”

1. Customer enters

2. Customer is seated
 3. Customer orders
 4. Waiter serves
 5. Customer pays
- Useful for **story understanding, planning, and natural language understanding**.
 - Handles **procedural knowledge** (how things happen).

Ontological Engineering

Definition:

- The process of **designing, building, and maintaining ontologies** for a domain.
- **Ontology**: A formal representation of **concepts, relationships, and constraints** in a domain.

Purpose:

- To allow **machines to understand domain knowledge** in a structured, shareable way.

Components of Ontology:

1. **Classes (Concepts)** – e.g., Animal, Vehicle
2. **Instances** – e.g., Lion, Car123
3. **Relations** – e.g., isA, partOf, hasColor
4. **Attributes/Properties** – e.g., age, color, speed
5. **Constraints / Rules** – e.g., Cars must have 4 wheels

Example (Simple Animal Ontology):

- Class: Animal
 - Subclass: Mammal
 - Subclass: Dog
 - Instance: Rover
- Relations:
 - Dog hasPetOwner Person
 - Dog hasColor Brown

Uses:

- Semantic Web (RDF, OWL)
- Knowledge Management Systems
- AI reasoning and planning

Categories and Objects in AI

In AI, **knowledge is often represented using categories and objects**. This is a common way to organize information for reasoning.

A. Categories (or Classes)

- A **category** is a group of objects that share common properties or features.
- Represents **general knowledge**.
- Also called a **concept** or **class**.

Example:

- Category: **Bird**
 - Common properties: has wings, can fly, lays eggs

B. Objects (or Instances)

- An **object** is a specific example of a category.
- Inherits properties from its category.

Example:

- Object: **Tweety**
 - Tweety is a **Bird**, so it has wings, can fly, lays eggs

REASONING SYSTEMS FOR CATEGORIES

1. Semantic Networks

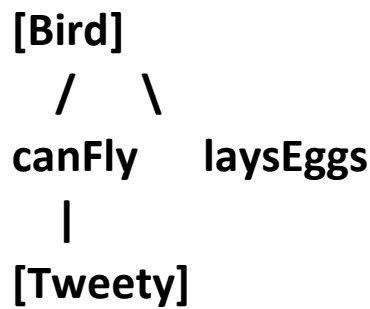
Definition:

A **semantic network** is a **graph-based knowledge representation** where:

- **Nodes** represent **concepts or objects**
- **Edges (links)** represent **relationships** between them

It's like a **visual map of knowledge**.

EXAMPLE:



- **Bird** is a category
- **Tweety** is an object (instance of Bird)
- Tweety inherits the properties **canFly** and **laysEggs**

Extensions:

- Can represent exceptions: e.g., [Penguin] → cannotFly

2. Description Logic (DL)

Definition:

Description Logic is a **formal knowledge representation language** used to describe:

- **Concepts (classes)**
- **Roles (relationships)**
- **Individuals (instances/objects)**

It is the **logical foundation of ontologies** (e.g., OWL in Semantic Web).

Key Components

1. **Concepts (C, D):** Classes or categories
 - Example: Bird, Penguin
 - 1.1 - Every penguin is a bird
2. **Roles (R):** Relationships between concepts
 - Example: hasPart, canEat
 - 1.1 - Every bird has at least one wing
3. **Individuals (a, b):** Instances of concepts
 - Example: Tweety, Pingu

- 1.1 -
 - Pingu : Penguin $\rightarrow$ Pingu is a Penguin, which is a Bird too
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